

# Quantum Gravitation

## Lecture 1

# Parameters of the Low-Energy Model

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

**■ Degrees of freedom**

We separate propagating polarizations from constrained components before quantization.

$$N_{SM} = N_g + N_{other} = 47 + 19 = 66$$

**■ Quantization**

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$p^\mu = (E/c, p)_{F01}$$

**■ Spectrum and response**

Poles, residues, and group velocity are read from the same quadratic operator.

$$L_{66} = \sum_b L_b$$

## ■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

**■ Example**

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

# Parameters of the Low-Energy Model

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

Reading: Low-Energy Parameter Reference, intro; Seo manuscript, ch. 1; F-01 normalization card